QUAID-E-AWAM UNIVER

CIENCE & TECHNOLOGY

Database Management Systems

Week 1, Day 5

MySQL—A RDBMS

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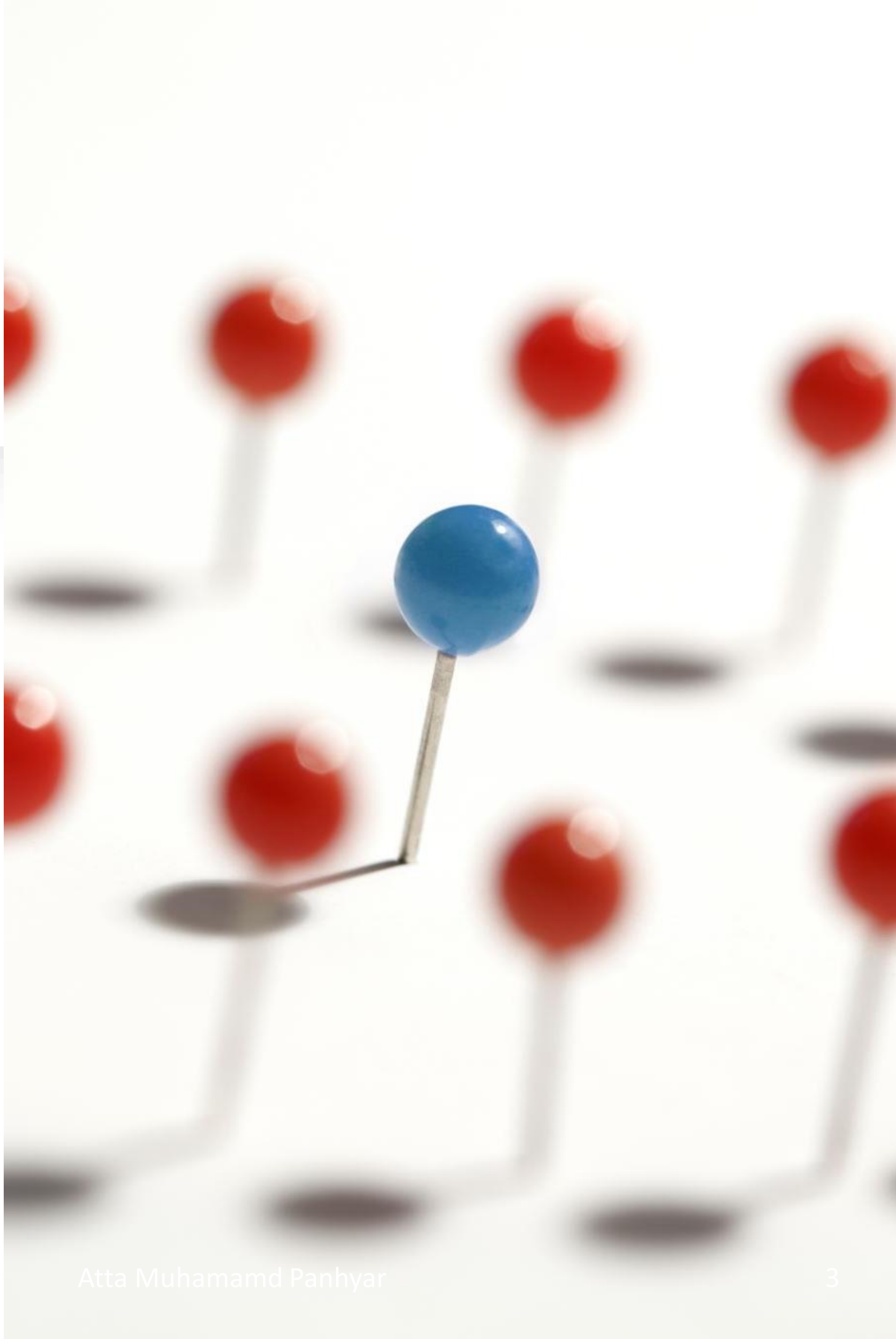
RECAP

- Order by
- Insert into
- Update …set
- Delete



LIKE Operator

- The LIKE operator is used in a WHERE clause to search for a specified pattern in a column.
- The percent sign (%) represents zero, one, or multiple characters
- The underscore sign (_) represents one, single character

Like operator

- **Example:**

```sql
SELECT * FROM Customers
WHERE CustomerName LIKE 'a%';
```

esult Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

CustomerID	CustomerName	ContactName	Address	City
2	Ana Trujillo Empa...	Ana Trujillo	Avda. de la Const...	México D.F.
3	Antonio Moreno ...	Antonio Moreno	Mataderos 2312	México D.F.
4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London
NULL	NULL	NULL	NULL	NULL

LIKE Operator	Description
WHERE CustomerName LIKE 'a%'	Finds any values that start with "a"
WHERE CustomerName LIKE '%a'	Finds any values that end with "a"
WHERE CustomerName LIKE '%or%'	Finds any values that have "or" in any position
WHERE CustomerName LIKE '_r%'	Finds any values that have "r" in the second position
WHERE CustomerName LIKE 'a_%'	Finds any values that start with "a" and are at least 2 characters in length
WHERE CustomerName LIKE 'a__%'	Finds any values that start with "a" and are at least 3 characters in length
WHERE ContactName LIKE 'a%o'	Finds any values that start with "a" and ends with "o"

Like operator …

IN Operator

- The IN operator allows you to specify multiple values in a WHERE clause.
- The IN operator is a shorthand for multiple OR conditions.

```sql
SELECT * FROM Customers
WHERE Country IN (SELECT Country FROM Suppliers);
```

esult Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

CustomerID	CustomerName	ContactName	Address
4	Around the Horn	Thomas Hardy	120 Hanover Sq.
5	Berglunds snabbk...	Christina Berglund	Berguvsvägen 8
6	Blauer See Delika...	Hanna Moos	Forsterstr. 57
7	Blondel père et fils	Frédérique Citeaux	24, place Kléber
8	Bólido Comidas p...	Martín Sommer	C/ Araquil, 67
9	Bon app''	Laurence Lebihans	12, rue des Bouc...

IN Operator

Aliases

Aliases are used to give a table, or a column in a table, a temporary name.

Aliases are often used to make column names more readable.

An alias only exists for the duration of that query.

An alias is created with the AS keyword.

Example 1

```sql
SELECT CustomerID AS ID, CustomerName AS Customer
FROM Customers;
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

ID	Customer
2	Ana Trujillo Empa...
3	Antonio Moreno ...
4	Around the Horn

Example 2

```sql
SELECT CustomerName, CONCAT_WS(', ', Address,
PostalCode, City, Country) AS Address
FROM Customers;
```

sult Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

CustomerName	Address
Ana Trujillo Empa...	Avda. de la Const...
Antonio Moreno ...	Mataderos 2312, ...

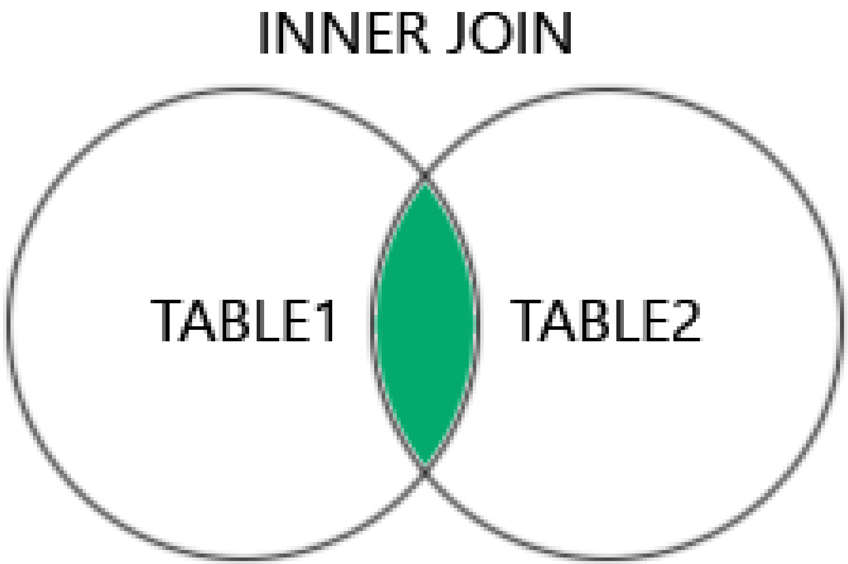
Example 3

```sql
SELECT o.OrderID, o.OrderDate, c.CustomerName
FROM Customers AS c, Orders AS o
WHERE c.CustomerName='Around the Horn' AND c.CustomerID=o.CustomerID
```

OrderID	OrderDate	CustomerName
10355	1996-11-15 00:0...	Around the Horn
10383	1996-12-16 00:0...	Around the Horn

JOINS

INNER JOIN



- The **INNER JOIN** keyword selects records that have matching values in both tables.

INNER JOIN –Example

```sql
SELECT O.OrderID, C.CustomerName
FROM Orders O
INNER JOIN Customers C ON O.CustomerID = C.CustomerID;
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

OrderID	CustomerName
10308	Ana Trujillo Empa...
10365	Antonio Moreno ...

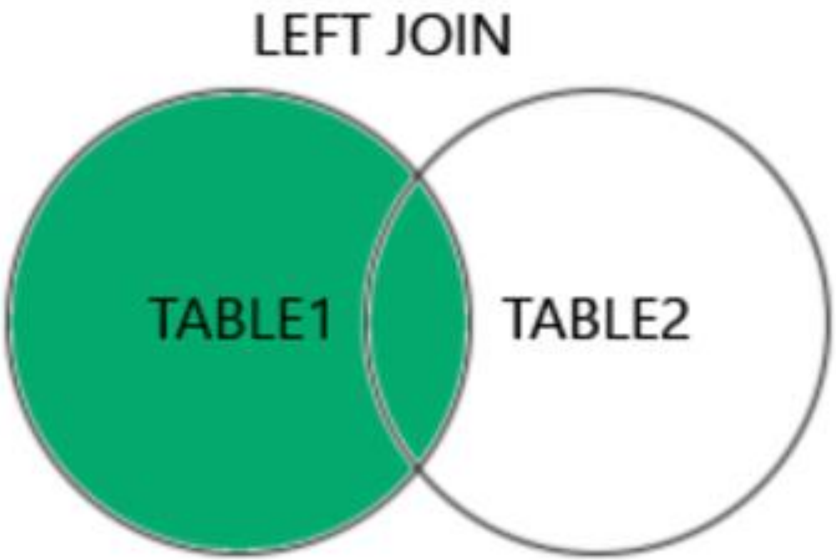
Example 2

```sql
SELECT o.OrderID, c.CustomerName, s.ShipperName
FROM ((Orders o
INNER JOIN Customers c ON o.CustomerID = c.CustomerID)
INNER JOIN Shippers s ON o.ShipperID = o.ShipperID);
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

OrderID	CustomerName	ShipperName
10308	Ana Trujillo Empareda...	Federal Shipping
10308	Ana Trujillo Empareda...	United Package
10308	Ana Trujillo Empareda...	Speedy Express
10365	Antonio Moreno Taqu...	Federal Shipping
10365	Antonio Moreno Taqu...	United Package

LEFT JOIN



- The LEFT JOIN keyword returns all records from the left table (table1), and the matching records (if any) from the right table (table2)

LEFT JOIN –Example

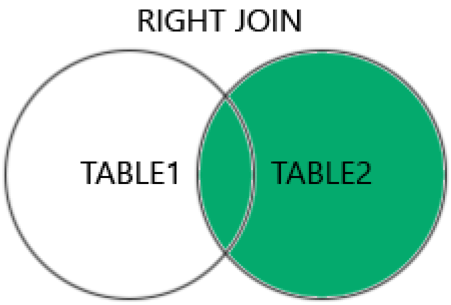
```sql
SELECT Customers.CustomerName, Orders.OrderID
FROM Customers
LEFT JOIN Orders ON Customers.CustomerID = Orders.CustomerID
ORDER BY Customers.CustomerName;
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

CustomerName	OrderID
Ana Trujillo Empareda...	10308
Antonio Moreno Taqu...	10365

RIGHT JOIN

- The RIGHT JOIN keyword returns all records from the right table (table2), and the matching records (if any) from the left table (table1).



RIGHT JOIN –Example

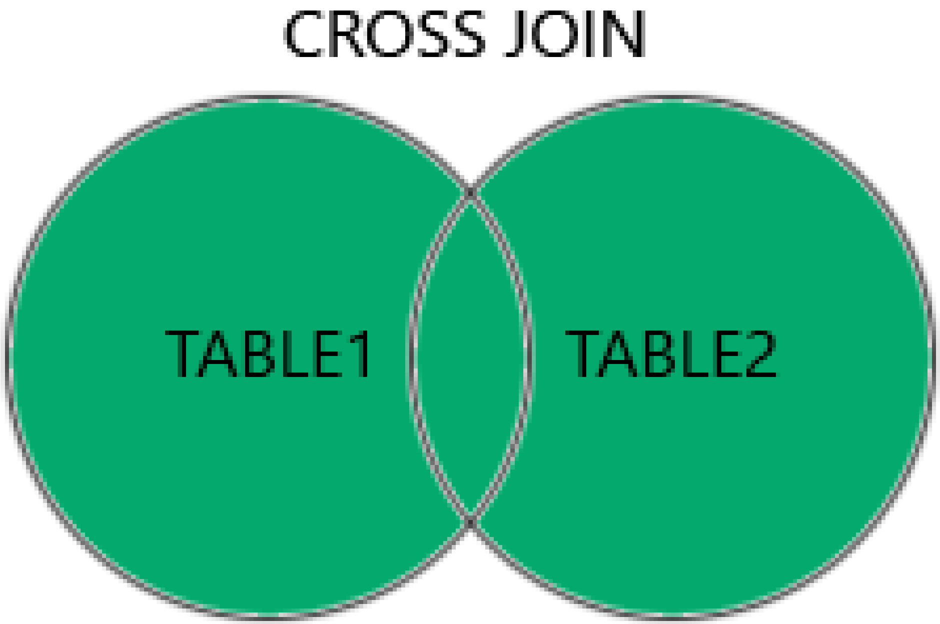
```sql
SELECT Orders.OrderID, Employees.LastName, Employees.FirstName
FROM Orders
RIGHT JOIN Employees ON Orders.EmployeeID = Employees.EmployeeID
ORDER BY Orders.OrderID;
```

sult Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

OrderID	LastName	FirstName
NULL	West	Adam
10248	Buchanan	Steven

CROSS JOIN

The CROSS JOIN keyword returns all records from both tables (table1 and table2).



CROSS JOIN –Example

```sql
SELECT Customers.CustomerName, Orders.OrderID
FROM Customers
CROSS JOIN Orders
```

sult Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

CustomerName	OrderID
Cardinal	10258
Cardinal	10258
Wolski	10258
Wilman Kala	10258

UNION

- The UNION operator is used to combine the result-set of two or more SELECT statements.
  - Every SELECT statement within UNION must have the same number of columns
  - The columns must also have similar data types
  - The columns in every SELECT statement must also be in the same order.

```sql
SELECT City FROM Customers
UNION
SELECT City FROM Suppliers
ORDER BY City;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

City
Aachen
Albuquerque
Anchorage
Ann Arbor

UNION

GROUP BY

- The GROUP BY statement groups rows that have the same values into summary rows, like "find the number of customers in each country".

- The GROUP BY statement is often used with aggregate functions (COUNT(), MAX(), MIN(), SUM(), AVG()) to group the result-set by one or more columns.

GROUP BY

```
SELECT  Country, COUNT(CustomerID) as 'No of Customers'
FROM Customers
GROUP BY Country;
```

ult Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Country	No of Customers
Mexico	5
UK	7

COMPLEX QUERY

```
SELECT Shippers.ShipperName, COUNT(Orders.OrderID) AS NumberOfOrders
FROM Orders
LEFT JOIN Shippers ON Orders.ShipperID = Shippers.ShipperID
GROUP BY ShipperName;
```

ult Grid | Filter Rows: | Export: | Wrap Cell Content:

ShipperName	NumberOfOrders
Speedy Express	54
United Package	74
Federal Shipping	68

HAVING Clause

- The HAVING clause was added to SQL because the WHERE keyword cannot be used with aggregate functions.

HAVING Clause

```sql
SELECT  Country, COUNT(CustomerID) as 'No of Customers'
FROM Customers
GROUP BY Country
HAVING COUNT(CustomerID) > 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Country	No of Customers
UK	7
Germany	10
France	11
Brazil	9

HAVING Clause

```sql
SELECT  Country, COUNT(CustomerID) as 'No of Customers'
FROM Customers
GROUP BY Country
HAVING COUNT(CustomerID) > 5
ORDER BY COUNT(CustomerID) DESC;
```

ult Grid | Filter Rows: | Export: | Wrap Cell Content:

Country	No of Customers
USA	13
France	11

HAVING CLAUSE

```sql
SELECT Employees.LastName, COUNT(Orders.OrderID) AS NumberOfOrders
FROM (Orders
INNER JOIN Employees ON Orders.EmployeeID = Employees.EmployeeID)
GROUP BY LastName
HAVING COUNT(Orders.OrderID) > 10;
```

ılt Grid Filter Rows: Export: Wrap Cell Content:

LastName	NumberOfOrders
Davolio	29
Fuller	20
Leverling	31

HAVING –One more

```sql
SELECT Employees.LastName, COUNT(Orders.OrderID) AS NumberOfOrders
FROM Orders
INNER JOIN Employees ON Orders.EmployeeID = Employees.EmployeeID
WHERE LastName = 'Davolio' OR LastName = 'Fuller'
GROUP BY LastName
HAVING COUNT(Orders.OrderID) > 25;
```

ult Grid | Filter Rows: | Export: | Wrap Cell Content:

LastName	NumberOfOrders
Davolio	29

Question(s)